

# ANDRY HUANG

## Fullstack Developer (.NET | Next.js | Node.js | AI/RAG)

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### PROFESSIONAL SUMMARY

Fullstack & AI Engineer with 6+ years of experience in system design, building, and scaling enterprise SaaS platforms, AI-driven applications, and cloud-integrated solutions. Expert in .NET Core, Next.js, NestJS, React, Node.js, and high-performance SQL optimization. Experienced working in remote international engineering teams (MRI Software - formerly Anacle Systems Ltd, Singapore), enterprise API integrations (DBS Bank, SharePoint), and building RAG & AI Agent workflows with OpenAI & Anthropic Claude APIs.

### TECHNICAL SKILLS

- **Frontend & Mobile:** Next.js, React, TypeScript, JavaScript, Tailwind CSS, Flutter, Android Native, Livewire, Alpine.js
- **Backend & APIs:** .NET Core / .NET Framework, C#, Node.js, NestJS, System Design & SaaS Architecture, Laravel (Livewire), Java Spring Boot, Python (FastAPI), REST & GraphQL APIs
- **AI & Automation:** RAG Architecture, OpenAI GPT-4o & Anthropic Claude APIs, Model Context Protocol (MCP), Autonomous AI Coding Agents, Vector DBs (pgvector), LangChain, BullMQ, n8n, Prompt Engineering
- **Databases & Cloud:** AWS (S3, EC2), Azure (Blob Storage, DevOps), PostgreSQL, MySQL, SQL Server, Oracle DB, Redis, Elasticsearch, Cloudflare R2
- **DevOps & Security:** GitHub Actions, Docker & Docker Compose, CI/CD Pipelines (Azure DevOps, GitLab CI, Jenkins), Linux Administration, Nginx, Fail2ban, SonarQube, SSH Security
- **Tools & Testing:** Playwright, Selenium, Git, Midtrans Payment Gateway, RajaOngkir, RFID Hardware Integration, Leaflet GIS Maps

### WORK EXPERIENCE

#### Fullstack Developer | MRI Software (formerly Anacle Systems Ltd) (Singapore (Remote)) | Nov 2022 - Mar 2026

- Engineered and maintained enterprise SaaS platforms (Anacle Simplicity & SMRT Tenant Management System), developing mall tender submission, tenant application processing, space leasing, and tenant operations using .NET Core, .NET Framework 4.8, Next.js, and NestJS.
- Built and deployed RAG architecture with OpenAI API and vector databases (pgvector) for automated report generation and semantic search.
- Refactored legacy .NET Framework 4.8 / SQL Server procedures and conducted Selenium automated UI testing across SaaS modules.

#### Key Achievements:

- ✓ **Go-Live SMRT Tenant Management:** Successfully delivered and launched the enterprise tenant management platform for SMRT Singapore, digitizing mall tender submissions, tenant applications, space leasing, and tenant operations.
- ✓ **Go-Live SharePoint Document Migration:** Engineered document migration service syncing 100,000+ enterprise files to SharePoint Graph API, AWS S3, and Azure Blob with RBAC & audit logging.
- ✓ **SonarQube Security & Quality Pipeline:** Spearheaded static code analysis across multiple internal enterprise projects, identifying and fixing security vulnerabilities, establishing automated CI/CD quality gates, and enforcing zero-vulnerability deployment standards.

#### Fullstack Developer | Software House (Surabaya, Indonesia) | Mar 2022 - Nov 2022

- Developed scalable web applications using Node.js and Next.js, optimizing high-concurrency SQL queries and backend services.
- Managed self-hosted Git servers, configured Elasticsearch infrastructure for log indexing, and upgraded enterprise Java runtime versions.

#### Key Achievements:

- ✓ **Database Cleanup & Data Recovery:** Successfully repaired and cleaned 10,000+ corrupted database records caused by legacy code logic bugs.
- ✓ **Incident & Ticket Reduction:** Resolved core architectural bottlenecks, eliminating recurring high-priority incident tickets.

#### Full-Stack Developer | PT Tjiwi Kimia (Indonesia) | Sep 2019 - Feb 2022

- Developed enterprise web solutions and internal reporting modules for pulp & paper manufacturing using .NET Core, C#, ASP Classic, and SQL Server.
- Built an automated WhatsApp Scheduler Bot using Python, Java, & Selenium for real-time production reporting and alert dispatches.
- Designed and deployed a native Android Barcode Scanner Mobile Application for real-time warehouse inventory stock-in / stock-out tracking.

- Developed an automated COVID-19 Employee Health Survey & Questionnaire web application for company-wide pandemic check-ins and contact tracing.
- Developed an automated lottery & raffle draw web application for Tjiwi Kimia's company anniversary celebration events.

#### Key Achievements:

- ✓ **Automated OEE & HWT Metrics Reporting:** Python & Java WhatsApp Bot auto-captured real-time production analytics, directly enabling plant operations to achieve target Overall Equipment Effectiveness (OEE) and Heavy Weight Target (HWT) machine performance.

## FEATURED PROJECTS & INNOVATIONS

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- **AI Portfolio Assistant & Interactive Chatbot** ([andryhuang.com](http://andryhuang.com)): Interactive RAG-powered portfolio chatbot allowing visitors and recruiters to query career history, tech stack, and project architecture in real time. Tech: Next.js, OpenAI API, Vector Embeddings, Tailwind CSS.
- **Qualiv | AI Recruitment SaaS Platform (Founder & Lead Architect)** ([qualiv.id](http://qualiv.id)): Built an end-to-end multi-tenant AI recruitment SaaS: LLM CV screening (PDF/Docx), candidate logic tests, real-time AI video & chat interview simulation, and scoring engine. Built with Next.js, NestJS, OpenAI GPT-4o API, PostgreSQL, Prisma, BullMQ + Redis Queue, Cloudflare R2, Midtrans.
- **OpenClaw | AI-Powered VPS Security & Ops Agent** ([Autonomous Agent](#)): Autonomous server security & monitoring agent deployed on remote Linux VPS nodes. Features LLM log anomaly diagnosis, real-time SSH brute-force detection & fail2ban auto-mitigation, and instant Telegram Bot alerts. Tech: Node.js, OpenAI API, Telegram Bot API, Fail2ban, Docker.
- **PT Nagamasban | AI Sales Assistant & Catalog** ([katalog.nagamasban.com](http://katalog.nagamasban.com)): AI recommendation engine and sales rep training platform for PT Nagamasban tire distributor in Surabaya, Indonesia. Matched tire specs with vehicle models and budget via OpenAI GPT-4o vector embeddings. Tech: Next.js, NestJS, OpenAI API, Vector Embeddings.
- **AI Stock Portfolio Manager** ([Personal AI Project](#)): Personal stock portfolio AI assistant with holdings tracking, financial report AI summarization, risk exposure calculator, and LLM analysis chatbot. Tech: Python FastAPI, Next.js, OpenAI GPT-4o, PostgreSQL, Lightweight Charts.
- **Retail & Enterprise Systems (Client Solutions)** ([Client Solutions](#)): Built custom enterprise systems for client businesses: (1) Cahaya Terang Tire ERP (Tire Store, Surabaya: Double-entry A/R & A/P Ledger), (2) Wijaya Mas Jewelry POS (Jewelry Store, Pluit Jakarta: RFID Hardware Scanner integration), (3) Knit and Cro E-Commerce (Yarn Store, Pluit Jakarta: knitandcro.com - Midtrans & RajaOngkir API), and (4) Tiga Putra Grocery POS (Grocery Store, Jombang: FIFO Expiry tracking). Tech: Laravel, Livewire, MySQL.
- **CG Home Sharing Platform (Community Initiative)** ([Volunteer Platform](#)): Airbnb-style Connect Group room booking platform for church community organizations featuring Leaflet GIS maps, surveyor inspection workflow, and SSO verification. Tech: Next.js, NestJS, PostgreSQL, Prisma, Leaflet GIS.
- **Interactive Quiz & Logic Test App (Community Initiative)** ([Volunteer Platform](#)): Real-time online quiz and logic testing platform built for community engagement, featuring real-time WebSocket scoring engines and participant performance analytics. Tech: Next.js, NestJS, PostgreSQL, WebSockets.

## EDUCATION

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**Bachelor of Science in Computer Science (S1 Teknik Informatika)** | University of Surabaya (UBAYA), Indonesia (2015 - 2019) | **GPA:** 3.60 / 4.00 (Cum Laude)

## CERTIFICATIONS

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- **Anthropic AI Certifications:** Building with Claude API, Model Context Protocol (MCP), Claude Code & AI Subagents (Anthropic, 2026)
- **AI Agents with Model Context Protocol:** Vanderbilt University (Coursera, 2026)
- **Software Architecture & Design of Modern Large Scale Systems:** Udemy (2026)

**Languages:** English (Professional Working Proficiency), Indonesian (Native)